

Hybrid Quantum–Classical Call-Center Staffing

Dual SLA/ASA Feasibility · Minimum Cost · PCE & CVaR-QAOA

Technical Presentation

Problem & scoring rule

Business goal

Match agents to forecasted demand across shifts so service quality holds and labor cost is minimized.

Official score

SLA $\geq 80\%$ and ASA ≤ 25 s $\Rightarrow$ lowest staffing + OT cost.

- Shifts, breaks, OT multipliers, skill/cover incidence $C_{s,t}$
- Interval demand $\lambda_t \rightarrow$ Erlang-C requirements $R_t^{\text{SLA}}, R_t^{\text{ASA}}$
- Plan accepted only after *true* Erlang simulation (not only linear R)

Unary variables: $x_{s,k} \in \{0, 1\}$, $n_s = \sum_k x_{s,k}$, $\text{cov}_t = \sum_s C_{s,t} n_s$

$$\begin{aligned} \min_n \quad & \sum_s c_s n_s \quad \text{s.t.} \quad \text{SLA}(n) \geq \text{SL}^*, \\ & \text{ASA}(n) \leq \text{ASA}^*, \\ & 0 \leq n_s \leq N_{\max}. \end{aligned}$$

QUBO (slack-free):

$$h(x) = \sum_s c_s n_s + P_{\text{SLA}} \sum_t (\text{cov}_t - R_t^{\text{SLA}})^2 + P_{\text{ASA}} \sum_t w_t (\text{cov}_t - R_t^{\text{ASA}})^2$$

→ Ising H ; no capacity slack qubits.

Solution architecture

- ① Synthetic history $\rightarrow$ forecast $\rightarrow$ Erlang $R^{\text{SLA}}, R^{\text{ASA}}$
- ② Build dual QUBO / Ising
- ③ **Quantum step (one of):**
 - Warm-start **CVaR-QAOA** (tail-focused sampling)
 - **PCE**: 12 vars $\rightarrow$ **5 qubits** via Z/ZZ correlators
- ④ Classical repair (coverage + ASA) + optional 1-flip LS
- ⑤ Hard Erlang dual gate $\rightarrow$ rank by cost

Classical ladder: greedy · multi-start LS · linear ILP · **exact enum** (ground truth).

What stands out

- **Dual metric**, not SLA-only — matches the scoring rule exactly
- **Honest baselines**: exact dual-feasible optimum, not only greedy
- **PCE compression**: same optimum at 5 qubits ($\sim 58\%$ fewer)
- **CVaR + warm-start**: low-energy tail + classical bitstring init
- **Slack-free QUBO**: avoids qubit blow-up from capacity slacks
- **Verified multi-seed**: dual-feasible 5/5; matches exact cost

Verified results (standard instance)

Method	Cost	Gap	Dual?	Qubits
Exact enumeration	\$1496	\$0	Yes	—
Multi-start LS	\$1496	\$0	Yes	—
PCE + 1-flip	\$1496	\$0	Yes	5
CVaR-QAOA hybrid	\$1496	\$0	Yes	12
Greedy	\$1672	+\$176	Yes	—
Linear ILP (CBC)	\$2016	+\$520	Yes	—

Exact plan: $n^* = [4, 3, 2]$ (Early/Mid/Late).

PCE and CVaR recover the **exact dual-feasible minimum cost**.

CVaR-QAOA

- 12 unary qubits
- Warm R_Y from greedy
- CVaR_α on energy tail
- Dual averaging optional
- Best-of-N + repair

PCE

- 12 vars $\mapsto$ 5 qubits
- Degree-2 Z/ZZ map
- Soft energy on $\langle P_i \rangle$
- Sign decode $\rightarrow$ bits
- 1-flip post-process

Both hit \$1496 dual-feasible; PCE is the scalability lever.

Classical baselines (fair comparison)

- **Exact enum**: all $5^3 = 125$ vectors — ground truth under scoring rule
- **Multi-start LS**: matches exact on all 5 seeds tested
- **Greedy**: dual-feasible but $\sim 12\%$ more expensive
- **Linear ILP**: over-staffs (hard R is conservative vs Erlang)

Judging guidance

Report gaps **vs exact**, not only vs greedy. Quantum matches exact; does not “beat” the true optimum.

Judging axes

Optimality Dual-feasible + cost = exact \$1496 on verified instance

Speed Classical exact/LS <10 ms; hybrid ~0.3–2 s (MPS sim)

Scalability PCE: 58% qubit cut; slack-free encoding; block decomposition path for enterprise size

Manager controls: cost / service / preference / resilience scenarios on the same model.

Summary

Delivered

End-to-end hybrid staffing solver with dual SLA/ASA gate, CVaR-QAOA, PCE (5 qubits), exact classical ground truth, multi-seed verification, math formulation, and notebooks.

Headline result

PCE + 1-flip and CVaR-QAOA recover the exact dual-feasible minimum cost (\$1496), matching multi-start local search and beating greedy by \$176 — with a clear, honest baseline ladder.